

Unsupervised Learning of Visual Features by Contrasting Cluster Assignments

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A. Summary

This paper proposes SwAV (Swapping Assignments between Views), an online clustering-based self-supervised method for visual representation learning. Instead of comparing features of different views directly as in contrastive learning, it assigns each view to learnable prototypes and predicts the cluster assignment of one view from the feature of another view of the same image. Together with multi-crop, a data augmentation strategy that adds low-resolution crops at little extra cost, SwAV achieves 75.3% top-1 accuracy on ImageNet linear evaluation with ResNet-50 and surpasses supervised pretraining on all considered transfer tasks.

B. Strengths

1. Memory efficient and flexible in batch size. SwAV avoids pairwise feature comparisons and requires neither a large memory bank nor a momentum encoder. It stores only about 3.8K features and works well with both large (4096) and small (256) batches, making it usable on GPUs with small VRAM.
2. Effective multi-crop augmentation. Adding several low-resolution crops to the two standard views increases the number of views at little extra memory cost, and consistently improves SimCLR, DeepCluster, and SeLa by 2-4% top-1 accuracy.
3. Truly online method. Unlike DeepCluster and SeLa, SwAV computes codes within each mini-batch. This allows it to scale to unlimited data, as shown by pretraining on 1B uncuration Instagram images.
4. Comprehensive experiments. SwAV is evaluated on ImageNet and multiple downstream tasks, surpassing supervised pretraining on all of them, with a fair ablation against contrastive methods.

C. Major Weaknesses

1. The benefit of the swapped mechanism is not isolated, and its claimed online advantage is not quantified. The headline result (75.3%, +4.2% over MoCov2) combines the SwAV loss with multi-crop, while most compared methods use two standard crops and different epochs/batch sizes. Under identical settings (Fig. 3), SwAV improves over SimCLR by 1.9%, but is on par with DeepCluster-v2 (70.1 vs. 70.2; 74.1 vs. 74.3; 75.3 vs. 75.2 in

the best setting). Since the paper itself interprets DeepCluster-v2 as a special case of swapping, the remaining argument for SwAV is that it is online. However, this advantage is argued only by construction and never measured: in the 1B uncuration experiment (Sec. 5.4), the only self-supervised baseline pretrained on the same data is SimCLR (the MoCo number is taken from [20] under a curated setting), and neither the wall-clock/memory overhead of DeepCluster-v2's offline clustering step nor a large-scale comparison with it is reported.

2. Key design choices (K and equipartition) are not analyzed in the main text. The number of prototypes is fixed ($\sim 3K$) without a stated principle, although it also determines the queue size in the small-batch setting ($\sim 3.8K$ features, since the partitioned set must be at least as large as K) and thus the claimed memory advantage. Equipartition prevents code collapse by construction, but the paper itself notes that collapse still occurs with a large ε (Sec. 3.1), so the constraint alone is not sufficient; yet no ablation (removing the constraint, varying ε or K) is shown. Moreover, equipartition forces uniform prototype usage over each batch (plus the feature queue), implicitly assuming a balanced concept distribution. Sec. 5.4 shows good aggregate accuracy on uncuration data, but reports no analysis of prototype usage or performance on rare concepts, so whether equipartition hurts long-tailed data remains unclear.
3. Compute claims are inconsistent, and the headline setting lacks a cost comparison. The abstract and introduction state that multi-crop adds no compute or memory overhead, whereas Sec. 3.2 acknowledges "a small increase in the compute cost". The former holds for $2 \times 160 + 4 \times 96$ ($\sim 0.88 \times$ the pixels of 2×224), but the headline $2 \times 224 + 6 \times 96$ setting processes $\sim 1.55 \times$ the pixels. Relative runtimes ($1.2 \times$ SimCLR, $1.4 \times$ MoCov2, attributed to "the additional back-propagation") are reported only for the small-batch $2 \times 160 + 4 \times 96$ setting (Sec. 4.3). For the headline 800-epoch, $2 \times 224 + 6 \times 96$ setting (Fig. 2), no cost comparison with baselines is given. A comparison at equal GPU-hours for this setting is needed to support the efficiency motivation.
4. (2026) Limited novelty and outdated baselines. Both technical contributions have since become

standard components: multi-crop is used in DINO, iBOT, and DINOv2, and Sinkhorn-Knopp-based balancing is used as the centering step in DINOv2. This adoption confirms the value of the ideas, but it also means that, submitted today, the paper would need to show what it adds beyond these frameworks. Yet it compares only with 2019–2020 methods, with no comparison against self-distillation (DINO, DINOv2), masked image modeling (MAE, iBOT), or language-supervised pre-training (CLIP), whose frozen features now far exceed 75.3% on ImageNet linear probing. Its main practical selling point, avoiding a momentum encoder and a large memory bank, is also less relevant, since the leading frameworks adopt a momentum teacher and still scale well on modern hardware.

5. (2026) Image-level objective and narrow evaluation. Codes are computed only from the globally pooled feature of each view, so the objective learns no patch-level or dense representation. Later work (e.g., iBOT and DINOv2) shows that adding patch-level objectives substantially improves localization-heavy tasks. The evaluation matches this limitation: it covers ImageNet linear probing and semi-supervised finetuning, three linear classification transfers, and two finetuned detection benchmarks, but no frozen-feature semantic segmentation, depth estimation, retrieval, or robustness benchmarks, which are now standard for evaluating general-purpose visual features.
6. (2026) Convolutional backbones and modest scale only. All results use the ResNet family (up to ~586M parameters and 1B images), whereas Vision Transformers are now the default backbone and self-supervised models are routinely scaled to billion-parameter ViTs. Although later work (DINO) shows that multi-crop can be adapted to ViTs via positional-embedding interpolation, this paper provides no evidence that the SwAV objective itself transfers to transformer backbones. Results on at least one transformer backbone, and scaling behavior beyond the ResNet family, would be necessary to support the method’s relevance today.

D. Minor Weaknesses

1. Mention the weaknesses that are easy to fix in a revision of the paper. For example:
2. Missing references that should be discussed but do not really change the novelty of the paper.

3. Occasional typographic errors.
4. Minor unclear points.
5. Mention here any minor suggestions, questions, corrections, etc. that can help the authors improve their paper.
6. (2026) Suppose this paper was submitted in 2026, what would be its minor weaknesses from today’s point of view?
7. (2026) In other words, this comment is based on current technological progress, not the time when the paper was originally published.

E. Justification

Balance the strengths and the weaknesses to explain why the paper should be accepted or not. Does the paper present sufficient knowledge advancement? Use your best judgment.

(2026) Suppose this paper was submitted in 2026, on what grounds would you recommend accepting or rejecting it from today’s point of view? In other words, this comment is based on current technological progress, not the time when the paper was originally published.

F. Final Rating

- 5: Strong Accept
- 4: Weak Accept
- 3: Borderline
- 2: Weak Reject
- 1: Strong Reject

Provide (2026) final rating as well.

This template follows the review format of CVPR 2025 [1] and [2].

G. Reflections

Share your reflections on this paper. What did you learn (technologies, writing, etc.)? How did it inspire you? Is it useful for your project or research? How might you apply it to your research? Just examples. Feel free to connect the paper to yourself in other ways.

References

- [1] CVPR 2025 Program Chairs. Cvpr 2025 reviewer guidelines. <https://cvpr.thecvf.com/Conferences/2025/ReviewerGuidelines>. 2025. 2
- [2] CVPR 2025 Program Chairs. How to be a good cvpr 2025 reviewer. https://docs.google.com/presentation/d/1fN6vhWAEELQQ67s7rqfR_JeQhp2cXwuD80m2qe3_g/edit?slide=id.p1#slide=id.p1. 2025. 2